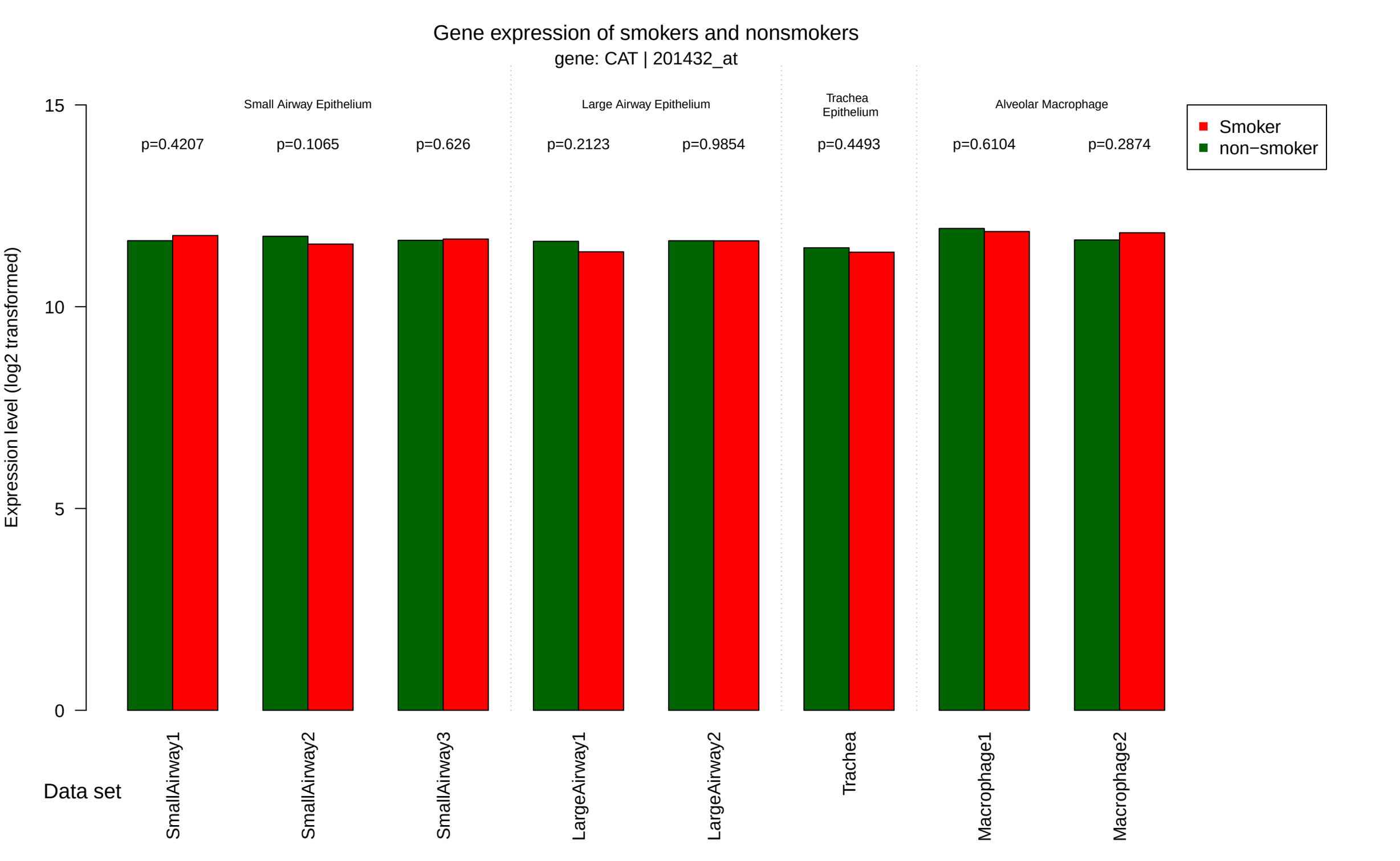
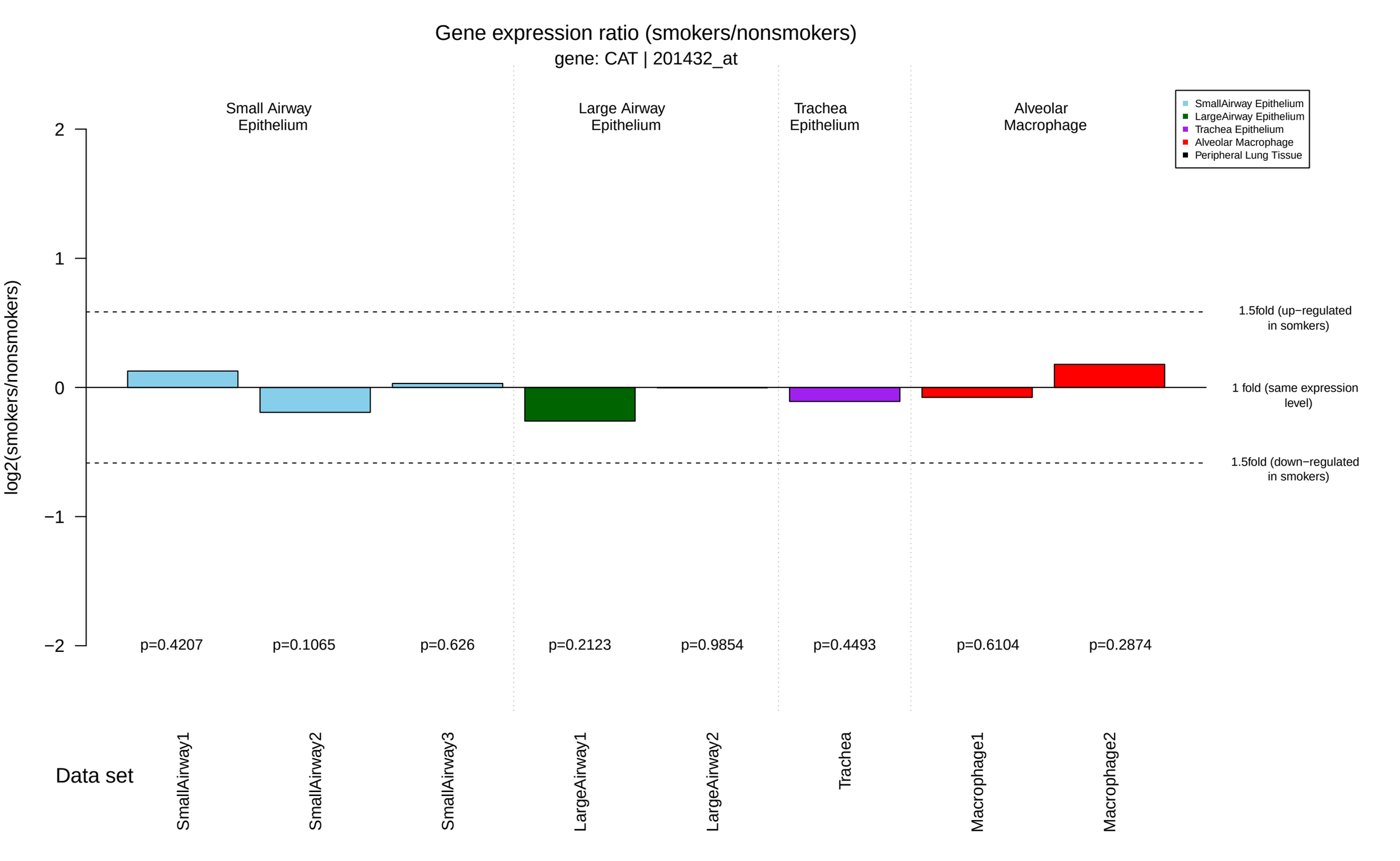
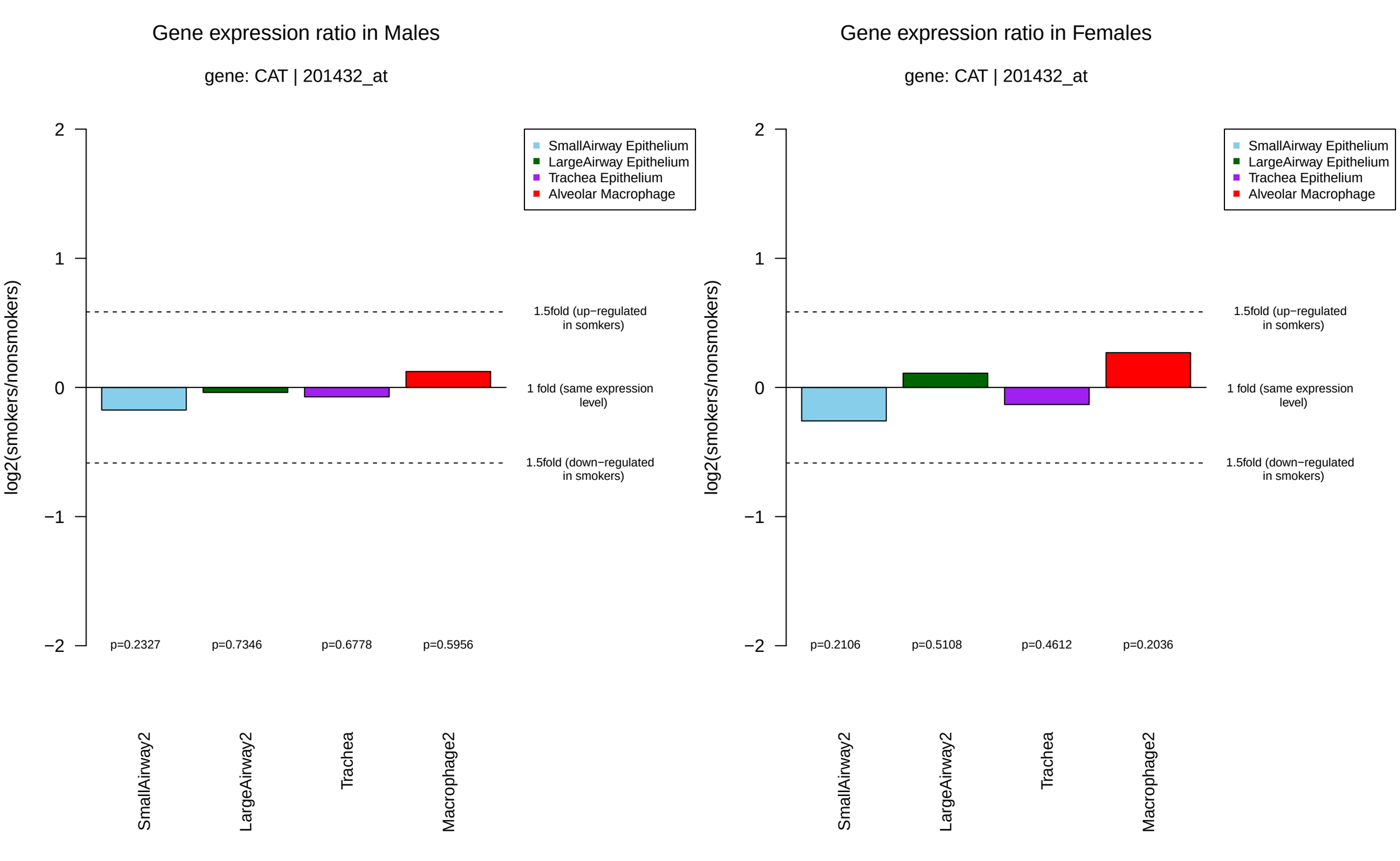
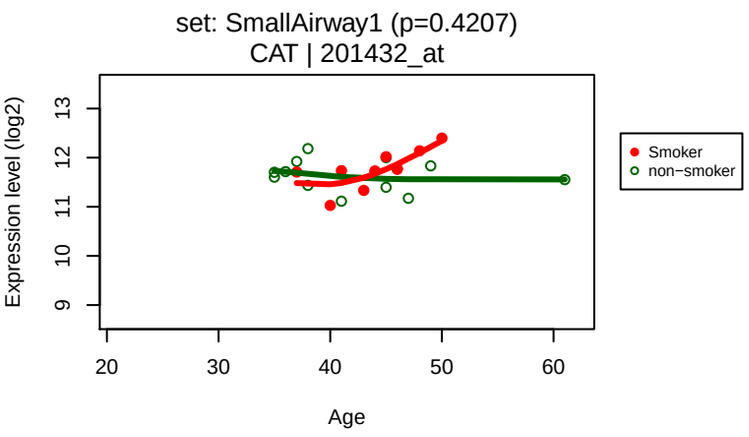
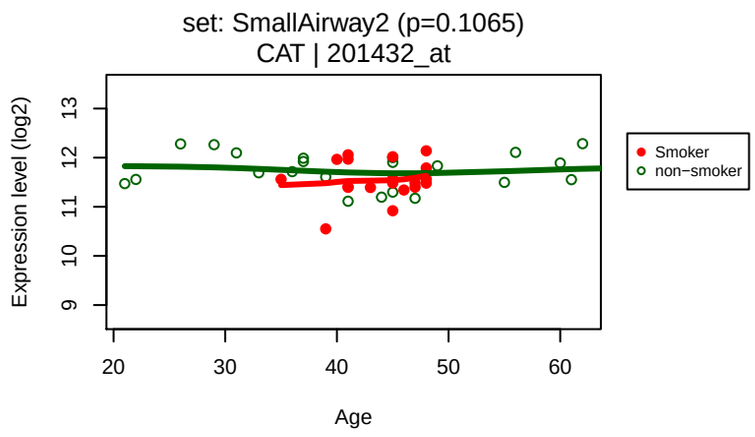
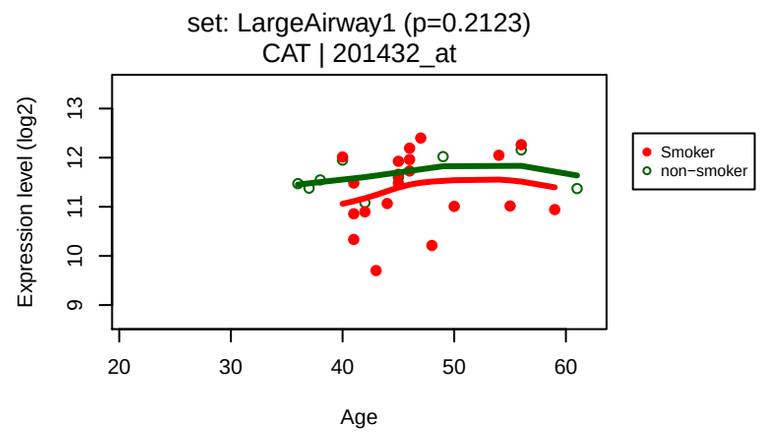

Gene expression of smokers and nonsmokers
gene: CAT | 201432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4207
p=0.1065
p=0.626
p=0.2123
p=0.9854
p=0.4493
p=0.6104
p=0.2874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

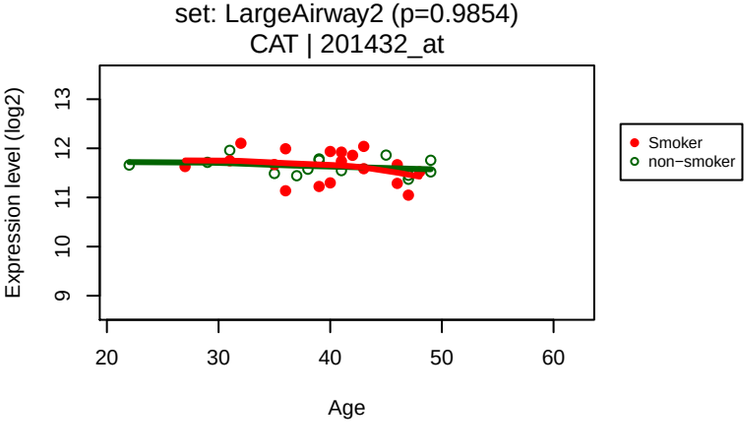

Gene expression ratio (smokers/nonsmokers)
gene: CAT | 201432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4207
p=0.1065
p=0.626
p=0.2123
p=0.9854
p=0.4493
p=0.6104
p=0.2874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

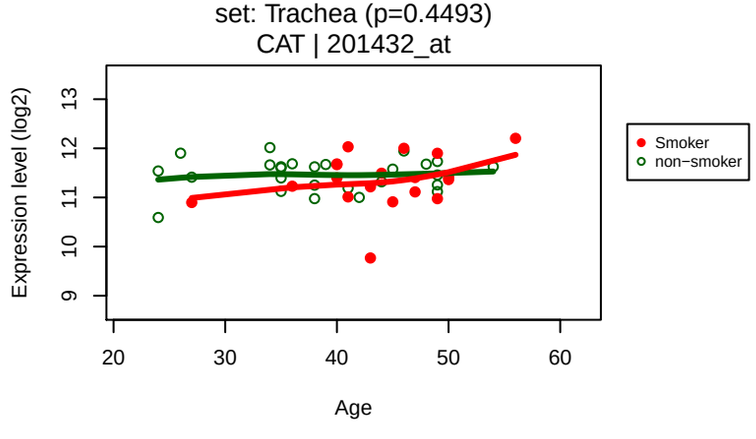

Gene expression ratio in Males
gene: CAT | 201432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2327
p=0.7346
p=0.6778
p=0.5956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAT | 201432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2106
p=0.5108
p=0.4612
p=0.2036
SmallAirway2
LargeAirway2
Trachea
Macrophage2

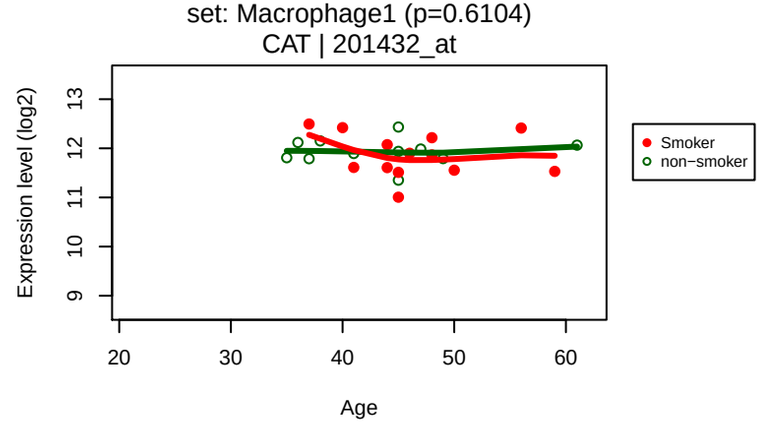

set: SmallAirway1 (p=0.4207)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker

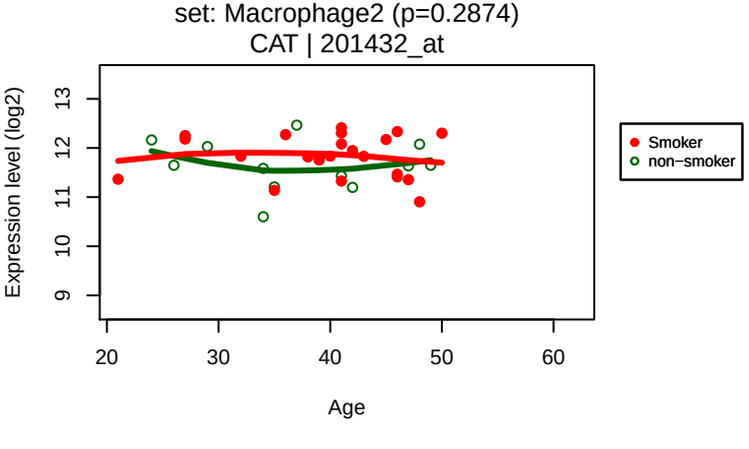

set: SmallAirway2 (p=0.1065)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker

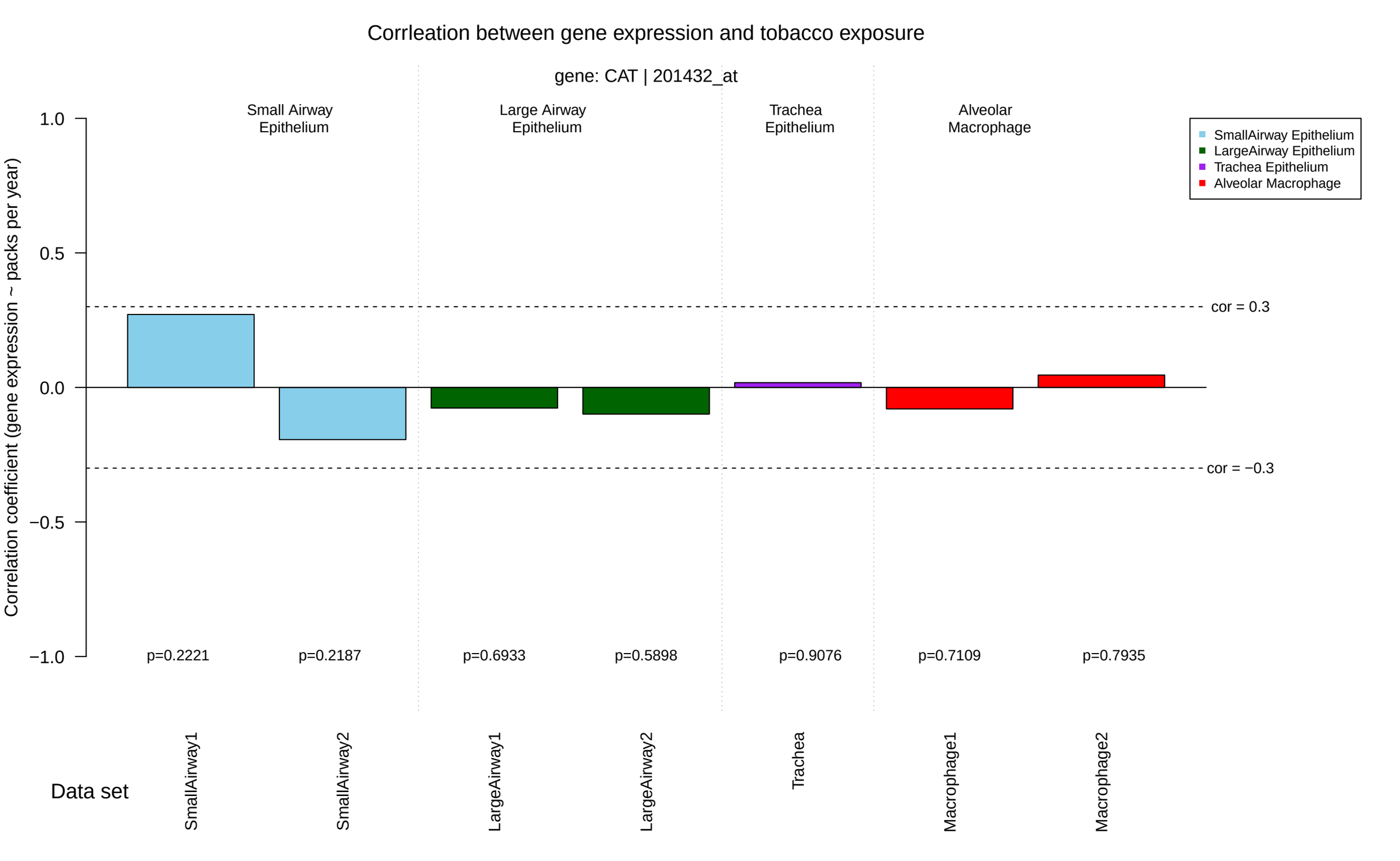

set: LargeAirway1 (p=0.2123)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9854)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4493)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6104)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2874)
CAT | 201432_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAT | 201432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2221
p=0.2187
p=0.6933
p=0.5898
p=0.9076
p=0.7109
p=0.7935
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2